

6. (2.75 points)

Given the following information:

<u>Cumulative Paid Loss (\$000)</u>					
Accident Year	12 Months	24 Months	36 Months	48 Months	60 Months
2012	500	750	950	1,050	1,100
2013	800	1,500	1,850	2,000	
2014	1,100	2,100	2,750		
2015	1,400	2,300			
2016	2,000				

- Age-to-age factors are determined by a volume weighted average of all years.
- There is no development beyond 60 months.
- Reserves are approximated by a lognormal distribution.
- The 90th percentile of the standard normal distribution is 1.28
- The accident year 2014 loss reserve estimate, R_3 , has a standard error of 52.33

a. (2 points)

Construct an 80% confidence interval for R_3 .

b. (0.75 point)

Construct a confidence interval for the accident year 2014 ultimate loss estimate using empirical limits.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 6										
TOTAL POINT VALUE: 2.75	LEARNING OBJECTIVE(S): A3									
SAMPLE ANSWERS										
Part a: 2 points										
<u>Sample 1</u> 36-48 LDF = 1.089 48-60 LDF = 1.048 -> 36-ult = 1.141 $R_3 = 2750 \times (1.141 - 1) = 387.75$ (\$000) Assume $se(R_3) = 52.33$ is also in (\$000) $\sigma^2 = \ln(1 + \frac{se(R_3)^2}{R_3^2}) = 0.018$ $\mu = \ln(R_3) - \sigma^2/2 = 5.951$ 80 C.I. ($e^{5.951 \pm 1.28 \sqrt{0.018}}$) -> (323.52, 456.11) (\$000)										
<u>Sample 2</u> $R_3 = R$ for AY 2014 LDF 36-48= (1050+2000)/(950+1850) = 1.0893 LDF 48-60 = (1100)/1050 = 1.0476 CLDF 36-ult = 1.1412 $R_3 = 2750K(1.1412-1) = 388180$ For lognormal distribution: $\sigma^2 = \ln(1 + (s.e.(R) / R)^2)$ = $\ln(1 + (52.33 / 388180)^2)$ = 0.000000018 80% CI for $R_3 = R_3 e^{(\pm 1.28(\sigma) - .5(\sigma^2))}$ = (388113, 388247)										
Part b: 0.75 point										
<table><tr><td></td><td>36-48</td><td>48-60</td></tr><tr><td>2012</td><td>1.105</td><td>1.048</td></tr><tr><td>2013</td><td>1.081</td><td></td></tr></table> Maximum bound is $(1.105 \times 1.048) \times 2750 = 3184.61$ Minimum bound is $(1.081 \times 1.048) \times 2750 = 3115.44$ Interval is (3115.44, 3184.61)			36-48	48-60	2012	1.105	1.048	2013	1.081	
	36-48	48-60								
2012	1.105	1.048								
2013	1.081									
EXAMINER'S REPORT										
Candidates were expected to demonstrate knowledge of the simulation of parameter percentiles										

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

and unpaid claims percentiles when models assume a distribution of residuals fit by MLE.

Part a

Candidates were expected to construct the 80% confidence interval for R_3 based on the lognormal distribution μ and σ .

Common errors include:

- Confusing the standard error of R_3 with σ
- Not providing a final numerical answer
- Using the wrong formula for the confidence interval
- Using the wrong LDF to calculate the confidence interval
- Not plugging in numbers to the confidence interval formula correctly
- Assuming a normal distribution for R_3
- Confusing R for C
- Using the reserves from all years as R , rather than just AY 2014

Part b

Candidates were expected to use the minimum and maximum age-to-age development factors to calculate a confidence interval for AY 2014 ultimate losses.

Common errors include:

- Providing the confidence interval for the reserves instead of the ultimates
- Using the wrong formula for the confidence interval
- Using the paid loss number from a year other than the year specified in the question
- Adding 2750 to the final answer from a